

PAUTREL Thibault

PhD

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Education

- 2024–2025 **Master 2, ENSAI**, M.Sc in data science
Machine Learning, Deep Learning, Time Series, NLP, Functional Data Analysis, Parallel computing
- 2018–2021 **PhD thesis**, *Université Rennes 1*, supervised by Prof. J. Angst and Prof. G.Poly
Some contributions to the universality of zeros of random trigonometric functions.
- 2017–2018 **Master 2 research-orientated**, *Université Rennes 1*, Stochastic processes and stoch. integration, (B)SDE, statistics
- 2018 **Internship**, *L.A.M.A. (Univ. Savoie-Mont Blanc)* under the supervision of Prof. Ph. Briand.
Backward and Forward Stochastic Diff. Equations with normal constraints
- 2016–2017 **Master 2 teaching-orientated**, *Université Rennes 1*, Preparation for the "agrégation" (highest competitive exam to recruit public teachers)
- 2015–2016 **Master 1 fundamental mathematics**, *Université Rennes 1*
- 2014–2015 **Mathematics degree**, *Université Rennes 1*
- 2012–2014 **Scientific Preparatory Classes MPSI-MP *** (option : computer science), *Lycée Chateaubriand*, Rennes

Work Experience

- 2023–2024 **Teacher in high school**, Lycée Bernard Palissy, (Gien)
Oral examiner, *weekly*, MPSI & BCPST1 Lycée Pothier (Orléans), PCSI & MP Lycée François 1er (Fontainebleau)
- 2024 **Centrale–Supélec exam assessor**, Written exam (MP), Oral exam (MP) : mathematics and computer science on Python
- 2022–2023 **Teacher in high school**, Lycée polyvalent Blaise Pascal, (Chateauroux)
- 2023 **Centrale–Supélec exam assessor**, Written exam (MP), Oral exam (MP) : mathematics and computer science on Python
- 2021–2022 **Teacher in secondary school**, Collège Saint Exupéry, (Villiers-le-Bel)
Oral examiner, *weekly*, BCPST2 Lycée Marcelin Berthelot (Saint-Maur-des-Fossés)
- 2018–2021 **Teaching assistant : supervision of practical work**, *64h/year*, Université Rennes 1, Mathematics department
- Analysis and probability course for undergraduate maths students
 - Probability course for first-year undergraduate biology students
 - Monitoring study-and-research dissertation for undergraduate students at ENS Rennes
- 2016–2018 **Oral examiner**, *weekly*, MP,MP * Lycée Chateaubriand (Rennes)

Research activities

Key-words **Nodal set, random trigonometric polynomials, dependent stationary Gaussian processes, spectral measure, Kac–Rice formula**

Conferences et seminars

- Invited speaker at "UniRandom" conference week, Rennes (2019, 2021 et 2023)
- Invited speaker at 'Journées des probabilités', Guidel, juin 2021
- Invited speaker at researchers' team weekly seminar, Rennes, novembre 2019

Publications

- T. Pautrel, *New asymptotics for the mean number of zeros of random trigonometric polynomials with strongly dependent Gaussian coefficients*, *Electro. Commun. Probab.*, 25 :13, 2020, Id/No 36 (2020)
- J. Angst, T. Pautrel et G. Poly, *Real zeros of random trigonometric polynomials with dependent coefficients*, *Trans. Amer. Math. Soc.*, 375, 7209-7260 (2022)

Exams

2017 **"Agrégation externe"**, *Probability and Statistics*, rank : 118

2012 **Scientific Baccalaureate**, Lycée Don Bosco (Mayenne)

Languages

French Native language

English Proficient CEFR C2

Certificate of Proficiency in English

Spanish Read, spoken and written

Computer skills

IT Tools Hadoop, NoSQL, Spark, Cloud Computing

Data Python, R, SQL

Unix Bash

Others $\text{\LaTeX}$, Office, OCaml, Scilab, Maple

Interests

Ski touring (France, Norway, Chilean Patagonia), alpine skiing

Running, cycling, hiking, swimming, tennis

Cooking